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S 430 FIELD FINISH PAINTING (FFP)

1. SCOPE OF WORK

1.1 The Bidder shall apply field finish paint all the buildings, equipment, structures, piping, and other accessories in accordance with the requirement of this specification and as listed in Schedule FFP herein related to the following:

1.1.1 Supply of all material, labor, scaffolding and equipment to perform the indicated work.

1.1.2 Field surface preparation.

1.1.3 Field touch-up painting.

1.1.4 Field touch-up for hot dipped galvanized surfaces.

1.1.5 Field prime coat painting.

1.1.6 Field intermediate and finish coat painting.

1.1.7 Color coding, identification and marking.

1.2 Finish coat painting for the structural steel shall be applied in the shop except connection parts. This specification S-430 shall be applied to these painting.

1.3 WORK EXCLUDED

This specification does not include field painting of the following materials and surfaces:

1.3.1 Copper and similar nonferrous metals.

- 1.3.2 Stainless steel.
- 1.3.3 Aluminum (except contact surfaces to prevent galvanic action)
- 1.3.4 Hot dipped galvanized grating.
- 1.3.5 Factory pre-finished surfaces except as stated in Schedule FFP herein.
- 1.3.6 Cast iron.
- 1.3.7 Plastic or plastic coated conduit.
- 1.3.8 Transits conduit.
- 1.3.9 Hot dipped galvanized cable trays and conduits.
- 1.3.10 Anchor bolts.
- 1.3.11 Milled surfaces.
- 1.3.12 Concrete floors, curbs
- 1.3.13 Floor and wall tiles.
- 1.3.14 Suspended ceilings.
- 1.3.15 Hardware and glazing.

2. INTENT OF SPECIFICATION

The intent of the requirements of this specifications for field painting is to ensure through and proper surface preparation, and careful and complete paint application to all exposed areas of surfaces specified to be field painted in this specification, and observing strictly the following procedure:

- 2.1 As the works are located in a tropical area and near the sea, the paint shall be of a type which can cope with tropical conditions and salty air environment.
- 2.2 The Bidder shall thoroughly clean surface of the applied prime coat where required and/or touch up the entire work, or a substantial portion thereof, before starting the intermediate coat of paint. However, the work shall proceed in a pattern that will keep to a minimum the necessity for recleaning the touch-up work already done or the intermediate coat already applied. Cleaning shall be done in such a manner as not to liberate dust. No intermediate coats shall be applied until the cleaning, prime coating and/or touch-up work have been approved by the Owner.

3. CODES AND STANDARDS

- 3.1 The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards”
- 3.2 The codes and standards of the following organizations extracted from the Part 4, subsection 2.5 “Codes and Standards” are specifically applicable to the design, application and testing of the work included in this Specification.

SSPC.....Steel Structures Painting Council, USA
JIS.....Japanese Industrial Standards

4. GENERAL REQUIREMENTS

4.1 Definitions

- 4.1.1 Contact Surfaces: Surfaces which are permanently in contact with other surfaces of the same material after shop assembly or field erection.(i.e. surfaces under column and beam, cover plates, faying surfaces, connecting angles, splice plates, bolts, nuts and washers)
- 4.1.2 Inaccessible Surfaces: Surfaces which are made inaccessible for painting (other than contact surfaces) after shop fabrication or field erection such as facing surfaces in an assembly composed of two angles back to back with a gusset plate between them.

- 4.1.3 DFT : Dry film thickness expressed in microns.
- 4.1.4 Outdoor : Areas of the Power Plant with or without roofs but without any walls
- 4.1.5 Indoor : Areas of the Power Plant enclosed with complete roof and walls
- 4.1.6 Piping : The term “piping” includes valves, pipelines, traps, strainers and all appurtenances in a piping system.

4.2 Safety Requirements

- 4.2.1 All safety measures and precautions required by the Owner and safety recommendations by material manufacturers, suppliers, etc. shall be carefully complied with during the entire course of the work.

4.3 Paint

- 4.3.1 The Bidder shall propose in his Bid two (2) or three (3) paint manufacturers. Paint manufacturer’s instructions: The manufacturers of the paints for the selected shall submit through Bidder printed instruction for the use of his materials. Where such instructions take exception to these specifications, the exception shall be submitted to the Owner in prior to commencement of painting works
- 4.3.2 Paint shall be delivered to the place of use in unbroken containers bearing manufacturer’s brand name and number.
- 4.3.3 Paint identification numbers provided by the selected paint manufacturer shall be used in ordering paint.
- 4.3.4 Paint shall not be mixed or stored in contaminated containers. All paint shall be thoroughly mixed before using at the shop or points of use.

4.4 Protection and Clean-Up for Field Work

- 4.4.1 The Bidder shall furnish the necessary clothes or other protection to fully protect all surfaces from paint drops, mist and other conditions due to execution of field prime coat, field touch-up, intermediate and finish painting work by the Bidder.
- 4.4.2 The Bidder shall exercise particular care to keep clean and unpainted all valve stems, motor shafts, and other such moving parts in order not to impair their free movement. Any paint deposited on these parts shall be carefully and completely removed.
- 4.4.3 All metal jackets on equipment and piping, cable trays, equipment nameplates, pressure gauges, instrument glasses, machined surfaces, electrical indicating devices etc. which are not to be painted shall be protected by masking to keep them free of paint.
- 4.4.4 Sufficient care shall be exercised by painters while painting around switches and controls so as to avoid tripping of circuits. It is imperative that all switches shall remain unmoved and undisturbed by painters.
- 4.4.5 All shipping tags, wires, strings and all other means of temporary or shipping identifications on surfaces to be painted shall be removed from such surfaces, but only after checking and receiving authorization from the Owner.
- 4.4.6 The Bidder shall clean all spots from surfaces which have been spotted with paint such as window glass, floors, walls, equipment etc. leaving the premises spot free to the complete satisfaction of the Owner.
- 4.4.7 Permanent names painted on equipment shall be protected so as to leave the names free of paints and legible.

4.5 Storage of Paint and Provision for Fire Fighting Equipment

- 4.5.1 A temporary building shall be provided by the Bidder for bulk storage of paint and paint material and for mixing of paint, thinner etc. as used from day to day. The building shall be properly ventilated and shall be adequately protected with firefighting equipment satisfactory to the Owner. The floor of

this building shall be kept clean at all times and oily rags, waste etc. shall be removed from this building at the end of each working day.

- 4.5.2 Adequate firefighting equipment, satisfactory to the Owner, shall also be provided by the Bidder at all locations where painting work is in progress.

4.6 Hoists and Scaffolds

The Bidder shall provide hoist and scaffolding so as not to interfere with the progress of other work or with the operation of the Plant. The Bidder's scaffolding, walkways etc. shall be complied with all applicable requirements of authorities having jurisdiction. The Bidder shall remove such scaffolding, walkways etc. at the termination of their use.

4.7 Miscellaneous Appurtenances

- 4.7.1 Control and regulating equipment linkages etc. shall be painted to match the equipment to which they are attached, unless otherwise directed by the Owner.
- 4.7.2 Small pipe, ducts and other small objects alongside or adjacent to large surfaces shall be painted to match the background color.
- 4.7.3 Brackets, hangers and other accessories shall be painted to match the color of support to which they are attached.

4.8 Technical Information

- 4.8.1 Where surfaces of equipment and structures have been prime coat painted in the shop, the Bidder shall review the compatibility of the field finish paint system with the shop prime paint prior to performing the work on them. Wherever the primer is not compatible to field finish paint or proper primer has not been used, the Bidder shall do the necessary work at no additional cost.

4.8.2 Field Touch-up Painting

- 4.8.2.1 Field touch-up painting where required shall consist of one coat of paint applied to the following surfaces after they have been properly cleaned as herein specified:
- a. Surfaces where the shop coat of paint has been marred, scratched or otherwise damaged due to shipping, handling, erection installation, weathering etc.
 - b. Heads of field bolts and nuts and adjacent surfaces left unpainted in the shop.
 - c. Surfaces of field welds and adjacent surfaces left unpainted in the shop.
 - d. Surfaces of any ferrous fasteners not otherwise protected.
- 4.8.2.2 Exposed fabrications, erection or shipping marks shall be cleaned off and the areas for touch-up shall be painted to match the adjacent surfaces.
- 4.8.2.3 Field touch-up painting shall consist of two coats for any of the above surfaces which have received two coats either in the shop or in the field prime coat painting.
- 4.8.2.4 For surfaces where blast cleaning and wash primer are specified touch-up painting shall include blast cleaning and applications of the wash primer before the touch –up coat, or coats, are applied.
- 4.8.2.5 Touch-up painting shall overlap the prime coat with not less than 25 mm wide all around to ensure continuity of coating.
- 4.8.3 Equipment and piping insulated and finished with aluminum jacketing shall not be painted. However, the color bands, follow direction arrows, names of the fluids for the piping and names of the equipment shall be indicated on the piping and equipment. Exposed hangers, rods, support and accessories shall be painted.

- 4.8.4 All bare (uninsulated) piping or outdoor use shall be painted, including piping in pits and trenches.
- 4.8.5 All hangers and supports shall be painted except sliding surfaces, springs and adjustable or movable parts.
- 4.8.6 All bare (uninsulated) equipment of indoor or outdoor use shall be painted, including motor base plates, stop valves and others.
- 4.8.7 Underground piping which is asphalt coated for corrosion protection, parts of asphalt coated pipe exposed above ground or above floors shall be painted.
In general, asphalt coating shall not be removed but shall be thoroughly cleaned by brushing, washing and/or chipping as necessary. Compatible primer and paints shall be used.
- 4.8.8 Surfaces of all structural piping, equipment etc. included in this Contact which will be made inaccessible for future painting after the erection of such items due to contact with other pipings, equipment, walls, roofs, ceilings etc., shall be field finish painted complete prior to such surfaces becoming inaccessible.

5. SPECIFIED SURFACE CLASSES

5.1 General

The specified surface classes and the general nature of the specified surface included in each class, shall conform to the requirements as specified herein.

5.2 Classes

The specified surface included in the work consist of the following four classes:

- 5.2.1 Class SSS (Specified Equipment Surfaces): Includes all exposed surfaces of structural steel, miscellaneous steel, auxiliary steel, galleries, handrails, exterior doors etc. as indicated in Schedule FFP.

5.2.2 Class SES (Specified Equipment Surfaces): Includes all exposed surfaces of the following, including all auxiliaries and accessories, as applicable:

- a. Mechanical equipment including tanks.
- b. Electrical equipment excluding electrical cubicles and indoor panels.
- c. Electrical fittings, junction boxes, outlet boxes, fasteners and all appurtenances.
- d. All other items of similar nature the foregoing as indicated in Schedule FFP or as required to properly complete the work.

5.2.3 Class SPS (Specified Piping Surfaces): Includes all exposed surfaces of the following, including valves, traps, strainers and all appurtenances, accessories, hangers, supports etc. as applicable:

- a. Mechanical piping.
- b. All other items of similar nature to the foregoing as indicated in Schedule FFP or as required to properly complete the work.

5.2.4 Class SBS (Specified Building Surfaces):
Includes all exposed surfaces of the following as applicable:

- a. Plastered surfaces
- b. Interior doors, rolling doors
- c. Ornamental handrails
- d. Wood work
- e. All other items of similar nature to the foregoing as indicated in Schedule FFP or as required to properly complete the work.

5.2.5 **Schedule FFP**

Schedule FFP lists the specified areas and surfaces, finish applied in the shop, field surfaces preparations system, field paint system and colors.

SCHEDULE FFP

ITEM No	SPECIFICATED AREAS AND SPEC- IFIED SURFACES	PRE- SENT FINISH	SURFACE PREPARA- TION RE- QUIRES (MINI- MUM)	PAINT SYS- TEM FILL IN BY BIDDER (NOTE-1)	COLORS (BY THE OWNER	
					INTERME- DIATE COAT	FINISH COAT
					(N O T E – 2)	
1.	CLASS SSS Specified Structural Surfaces					
1.1	All structural steel, auxiliary steel, mis- cellaneous steel, gal- leries, trolleys, stairs, ladders, cages, handrails, exterior doors etc. as furnished by the Bidder					
1.1.1	Indoor	Shop prime painted	SSPC-SP3		Later	Later
1.1.2	Indoor	Bare met- al	SSPC -SP4		Later	Later
1.1.3	Outdoor	Shop prime painted	SSPC -SP3		Later	Later
1.1.4	Outdoor	Bare met- al	SSPC -SP4		Later	Later
1.2	All exposed steel surfaces, supports etc.	Shop prime painted	SSPC -SP3		Later	Later
2.	CLASSESES Specified Equipment Surfaces:					
2.1	All bare mechanical equipment, pumps, ventilation etc, in- cluded in the Con- tract.					

SCHEDULE FFP

ITEM No	SPECIFICATED AREAS AND SPECIFIED SURFACES	PRE-SENT FINISH	SURFACE PREPARATION REQUIRES (MINIMUM)	PAINT SYSTEM FILL IN BY BIDDER (NOTE-1)	COLORS (BY THE OWNER)	
					INTERMEDIATE COAT	FINISH COAT
					(N O T E – 2)	
2.1.1	Indoor	Shop prime painted	SSPC -SP3	None	Later	Later
2.1.2	Outdoor	Shop prime painted	SSPC -SP3		Later	Later
2.2	All jacketed mechanical equipment, pump etc. included in the Contract.	Shop prime painted	None		None	None
2.3	All bare carbon steel tanks, vessels included with the Contract					
2.3.1	Indoor	Shop prime painted	SSPC -SP3		Later	Later
2.3.2	Outdoor	Shop prime painted	SSPC -SP3		Later	Later
2.4	All electrical equipment, motors, junction and outlet with boxes Included in the Contract	-	-			
2.4.1	Indoor	Shop prime painted	SSPC -SP3		Later	Later
2.4.2	outdoor	Shop prime painted	SSPC -SP3		Later	Later

SCHEDULE FFP

ITEM No	SPECIFICATED AREAS AND SPEC- IFIED SURFACES	PRE- SENT FINISH	SURFACE PREPARA- TION RE- QUIRES (MINI- MUM)	PAINT SYS- TEM FILL IN BY BIDDER (NOTE-1)	COLORS (BY THE OWNER	
					INTERME- Diate COAT	FINISH COAT
					(N O T E – 2)	
2.5	Skids all equipment including exposed anchor bolts included in the Contract					
2.5.1	Indoor	Shop prime painted	SSPC -SP3		Later	Later
2.5.2	Outdoor	Shop prime painted	SSPC -SP3		Later	Later
2.6	All electrical conduits included in the Contract					
2.6.1	Indoor	Galva-nized	SSPC -SP3		Later	Later
2.6.2	Outdoor	Galva-nized	SSPC -SP3		Later	Later
3.	CLSS SPS (Specified Piping Sur-faces)					
3.1	All bare steel piping included in the Contract	Bare metal	SSPC -SP4	None	Later	Later
3.1.1	Indoor	Bare metal	SSPC -SP4		Later	Later
3.1.2	Outdoor	Bare metal	None		None	None
3.2	All insulated and jacketed steel piping both indoor and outdoor included in the Contract.					
3.3	All valves including hand wheel and connection bolts included in the Contract					

SCHEDULE FFP

ITEM No	SPECIFICATED AREAS AND SPECIFIED SURFACES	PRE-SENT FINISH	SURFACE PREPARATION REQUIRES (MINIMUM)	PAINT SYSTEM FILL IN BY BIDDER (NOTE-1)	COLORS (BY THE OWNER)	
					INTERMEDIATE COAT	FINISH COAT
					(NOTE - 2)	
3.1.1	Indoor	Bare metal	SSPC -SP4		Later	Later
3.3.2	Indoor	Bare metal	SSPC -SP3		Later	Later
3.3.3	Outdoor	Bare metal	SSPC -SP4		Later	Later
3.3.4	Outdoor	Shop prime painted	SSPC -SP3		Later	Later
3.4	All pipe hangers, duct hangers included in the contract					
3.4.1	Indoor	Shop prime painted	SSPC -SP3		Later	Later
3.4.2	Outdoor	Shop prime painted	SSPC -SP3		Later	Later
4.	CLASS SBS (Specified Building Surface)				Later	Later
4.1	All Plastered surfaces.	Bare	Per paint manufacturer's standard		Later	Later
4.2	All interior doors and rolling doors	Shop primed	Per paint manufacturer's standard	Varnish	Later	Later
4.3	All wood work	Bare	Per paint manufacturer's standard Anodized	-	Later	Later
4.4.	Aluminum windows					

Notes:

1. The Bidder shall fill up paint system in this column based on the paint system indicated in table S-430-1 (Paint System for Steel Surfaces) and Table S-430-2 (Paint System for Building Surfaces) as applicable.
2. The Owner will select the color of the finish coat after the award of the Contract at no extra cost to the Owner.
3. Bare metal surfaces shall receive shop and field coat paints as indicated in Table S-430-1 (Items 1, 2, 3, & 4)
4. Field touch up of primed metal surfaces shall be done using field coat paints as indicated in Table S-430-1 (items 3& 4).
5. Field touch up of galvanized steel surfaces shall be done using paint system 420-E in Table S-420-2, Specification S-420 “Prime Coat and Shop Finish Painting”.
6. Do not paint thiokol base or silicone base rubber sealant caulking. Paint all other caulking with same paint as adjacent or surrounding areas.
7. Safety color painting shall be approved by the Owner.

6. PAINTING WORK SYSTEM

- 6.1 Tables S-420-1 and S-430-2 show the paint systems suggested by the Owner, for both steel and building surfaces.
- 6.2 The Bidder shall study the paint system to be compatible with the material to be coated, the temperature of the structures/equipment/piping surfaces, service conditions and what can cope with the tropical conditions and salty air environment.

6.3 The Bidder shall submit his proposed painting system and additional surface preparation requirements complete on Schedule FFP herein for Owner's review and approval.

6.4 The Bidder shall provide a minimum 5 year paint performance guarantee against any potential failure of the paint system and discoloration. This guarantee shall be properly signed and submitted to the Owner.

7. DETAILS OF PAINTING WORK SYSTEMS

7.1 Surface Preparation System (Sspc Sp1~Sp3)

Shop primed metal or galvanized metal shall be as follows:

7.1.1 System Description

This system shall include preparations of all specified ferrous metal surfaces that will receive shop prime painted, or galvanized, and field touch-up and/or a field finish coat.

7.1.2 Surface Preparation

All surfaces shall be cleaned of rust, dirt, dust or other foreign materials. Use System SSPC-SP1, Solvent Cleaning, and SSPC-SP2, Hand Tool Cleaning, and SSPC-SP3, Power Tool Cleaning, if required.

7.1.3 Surface Appearance after Surface Preparation

The surface after surface preparation done shall be at least as good as indicated by preparation grade photo BSt3 and CSt3.

7.2 Surface Preparation System Sspc-Sp4

Bare metal to be coated or painted shall be as follows:

7.2.1 System Description

This system shall include preparation of any specified ferrous metal surfaces that have not been shop prime painted but shall be field prime coated or painted.

7.2.2 Surface preparation: Methods SSPC-SP1, Solvent Cleaning, SSPC-SP2, Hand Tool Cleaning, and SSPC-SP3, Power Tool Cleaning, SSPC-SP10, Near White Metal Blast Cleaning, where required.

7.2.3 Surface appearance after surface preparation: to be at least as good as indicated by preparation grade photo BSa2 ½ and CSa2 ½ of SSPC-Vis1.

7.3 Surface Preparation for Concrete and Plaster Surfaces

7.3.1 Concrete and plaster surfaces shall have ages at least 30 days prior to starting preparation for painting.

7.3.2 Surfaces shall be completely free from oil, grease lime or other contaminants.

7.3.3 Cracks, voids, etc shall be filled with spackling compound and sanded or rubbed smooth.

7.3.4 All rough spots shall be sanded smooth.

7.4 Surfaces Preparation for Wood Surface

7.4.1 Surfaces shall be sanded smooth.

7.4.2 All knots and sappy streaks shall be carefully coated with shellac before painting.

7.4.3 Suitable primer shall be used.

7.4.4 Holes, splits, scratches etc. shall be filled with spackling compound after primer has been applied and the affected areas again sanded smooth.

7.5 Surface Preparation for Galvanized Surfaces

7.5.1 Surface shall first be prepared by solvent cleaning.

- 7.5.2 Following the solvent cleaning, the surface shall receive one (1) wash coat pretreatment in accordance with the applicable requirements of SSPC Pretreatment Specification No. 3 “Basic Zinc Chromate-Vinyl Butyral Washcoat”, or shall receive a prime coat as specified.

7.6 Metal Patching

If it is found necessary to patch or fill scratches or gauges in equipment metal housing or motors, etc., a metal putty shall be used. Metal putty shall be applied in strict accordance with manufacturer’s direction, and shall be level and smooth after application.

8. COLOR CODING AND/OR IDENTIFICATION AND MARKING

- 8.1 Color band for jacketed pipes, flow direction arrows and names of fluids shall be located on piping system immediately adjacent to all operating accessories such as valves, regulators, strainers and other devices, including points where pipes branch or pass through walls, floors or partitions.

- 8.2 Color of the bare and jacketed pipes shall be as indicated in Table S-430-3 as applicable.

- 8.3 The final location of color bands, arrow marks and names shall be adjusted at site according to the direction of the Owner, taking the actual service operation into consideration.

- 8.4 Major equipment and auxiliaries shall be indicated on the surface of tank, pump or motor bodies etc.

8.5 Identification Names (Legends) and Arrows

- 8.5.1 Legends and arrows shall be painted on using stencils unless self-sticking legend marking tape is used.

- 8.5.2 legends and arrows shall be stenciled on at locations as follows:

- 8.5.2.1 At all valves

- 8.5.2.2 At intermediate points, approximately 8 m apart, on long pipe runs.
- 8.5.2.3 Legends shall be in letters of style and arrows shall be of size and shape as approved, for pipe lines 75 mm diameter or larger (this shall include insulation where used). For smaller pipe lines the legends and arrows shall be sized as large as each line will permit.
- 8.5.2.4 Paint for legends and arrows shall be Black Pipe Enamel unless otherwise indicated.

9. RESTRICTIONS DURING PAINTING

- 9.1 Applications of paints shall conform to the applicable requirements of SSPC Paint Application Specifications No. 1, "Shop, Field and Maintenance Painting" and to the requirements herein specified.
- 9.2 Paint shall not be applied to surfaces which are at a temperature of over 50°C unless the paint is specifically formulated for applications at the proposed temperature. When a surface is painted in hot weather, precautions shall be taken to ensure that the specified dry micron thickness of paint is obtained.

9.3 Moisture And Humidity Restrictions

- 9.3.1 Paint shall not be applied in rain, fog or mist or when the relative humidity is such as to cause condensation on metal surfaces due to variation in metal temperatures and surrounding air temperatures.
- 9.3.2 All surfaces shall be perfectly dry before and while being painted.

10. METHOD OF APPLICATION

- 10.1 Application of the paint shall primarily be made with proper size and grade of brushes except where spray equipment or rollers can be used to advantage.
- 10.2 Spray equipment, if used, shall be accurately and fully controlled at all times so as to prevent damage to adjacent surfaces. If, in the opinion of the

Owner, the overspray during paint spraying operations is causing damage to plant equipment the use of spray equipment will be restricted to those areas(if any) where, in the opinion of the Owner, damage to plant equipment will not occur.

- 10.3 The Bidder shall solvent clean or sand and smooth touch-up prime coats and undercoat paint on all equipment and on other readily visible surfaces to secure a smooth workmanlike finish coat.

11. INSPECTION

- 11.1 All work covered by this specification will be subjected to inspection by the Owner, and any work found not in accordance with the specification shall be redone to the satisfaction of the Owner.
- 11.2 The Owner may make random spot checks of materials, equipment etc. which have been painted by removing small swatches of paint to bare metal with acetone and examining the surface with a 10 power magnifying glass to observe the absence or presence of rust under the prime coat. The presence of rust or rust stains under such swatches shall be sufficient cause for rejecting the Work, and the Bidder shall prepare and recoat the Work in the field at his own expense.
- 11.3 Dry and wet film thickness may be determined by the use of gauges. The gauges on procedure shall be in accordance with SSPC PA-2.
- 11.4 To check the amount of thinner added the viscosity of paints in the cans may be determined by viscosity cups and compared to values obtained from the manufacturer.
- 11.5 The continuity of paint film on metal surfaces may be checked by a low voltage detector. Inadequately coated areas shall be marked with chalk. The Owner may consider the entire area defective if five or more deficiencies are found in an area of 10 m².
- 11.6 The surface temperature may be determined by the use of a surface thermometer.

- 11.7 Paint shall have sufficient adhesion to prevent hardened coats being lifted as a layer when the blade of a putty knife is inserted under it. Intercoat adhesion shall be such that two or more hardened coats shall not be delaminated when tested with a sharp knife.

- 11.8 All equipment necessary for quality control testing shall be provided by the Bidder when requested by the Owner.

12. SAMPLES

After award of Contract, but before ordering materials, Bidder shall submit for review of Owner all of the following samples:

- 12.1 Color Panels: Approximately 30 cm x 30 cm in size. Submit panels for each finish color to be used, including colors for all undercoats.
- 12.2 Color Chips: manufacturer's standard, but not less than approximately 25 mm x 50 mm in size. Submit chips for each finish color to be used including colors for all undercoats.
- 12.3 Self-sticking Tape: Samples of each size of self-sticking marking tape proposed for use.

13. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

- 13.1 Quality assurance/quality control shall conform to the requirements of Part 4, subsection "Quality Assurance/Quality Control".
- 13.2 The quality control procedures and documentation shall include but not be limited to the following :
- 13.2.1 Packaging and shipping.
- 13.2.2 Field cleaning and painting procedures.

TABLE S-430-1
PAINT SYSTEM FOR STEEL SURFACES

ITEM	COAT-TYPE-DFT	PAINT SYSTEM
1. SHOP PAINT	Shop-Prime Coat: (inorganic zinc 85 to 89 % zinc by weight) 50 to 75 microns	
2. SHOP PAINT	Shop-Intermediate Coat: (two component polyamide epoxy carboline) 125 to 175 microns	
3. FIELD PAINT	Primer-Intermediate Coat: (Touch-up and Total Coating) (aluminum-epoxy-mastic) 125 to 175 microns	
4. FIELD PAINT	Field Finish Coat: (polyurethane) 50 to 75 microns	

1. The Bidder shall propose paint system and shall fill in column above or any additional columns if required.
2. All paints shall meet the performance requirements.
3. The use of all components of the same manufacturer's paint system is mandatory.
4. Owner will select the colour scheme of field finish painting work to complement the building colour at a late date after the award of the Contract at no extra cost to Owner.

TABLE S-430-2
PAINT SYSTEM FOR STEEL SURFACES

ITEM	COAT-TYPE-DFT	PAINT SYSTEM
1.	One Latex Sealer Coat 12,5 to 25 microns	
2.	Intermediate Coat: Latex Flat Finish 25 to 50 microns	
3.	Finish Coat Latex Flat Finish (25 to 50 microns)	

ADD1
-IIG
-013

1. The Bidder shall propose paint system and shall fill in the column or any additional columns as required.
2. All paint shall meet the performance requirements.
3. The use of all component of the same manufacturer's paint system is mandatory.
4. The Owner will select the colour scheme of field finish painting work to complement the building colour at a later date after the award of the Contract at no extra cost to Owner.

TABLE S-430-3 COLOR SYSTEM FOR PIPING & TANKS

No.	Fluid Name	Color For Pipe / Tank
I	STEAM:	RED:
a.	Wet steam	Silver with red and green band
b.	Superheated steam	Silver with red and white band
c.	Exhaust steam	Silver with red and black band
II	WATER	GREEN:
a.	Feed Water	Green with red band
b.	Condensate water	Green with yellow band
c.	Sea water cooling	Green with yellow band
III	FUEL & LUBE OIL	BROWN:
a.	HSD & 1 DO	Brown with yellow band
b.	Residual	Brown with black band
c	Gas	Yellow
d	Lube oil:	
d.1	Inlet	Brown with white band
d.2	Outlet	Brown with red band
IV	AIR:	BLUE:
a.	Atomizing air	Blue with brown band
b.	Control air	Blue
V	WATER TREATMENT:	
a.	Acid (HCL)	Orange
b.	Alkali (NaOH)	White
VI	VACUUM	GREY
VII	TANK	Silver
	a) HSD	Brown with yellow band
	b) Water	Green with red band

NOTE:

Any color coding and/or identification at equipment, system and appurtenances not identified in this table, but within Bidder's scope work, will be finalized later by the Owner.

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